Regime-Adaptive Distribution-Free Calibration for Financial Tail Risk

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Abstract

We propose a model-agnostic calibration layer for one-sided tail-risk forecasts that provides finite-sample coverage guarantees under heavy tails, temporal dependence, and regime shift. Given any base Value-at-Risk (VaR) forecaster, our method computes calibration residuals, weights them by both exponential time-decay and a regime-similarity kernel derived from observable market features, and inflates the resulting weighted quantile by a correction term derived from the Dvoretzky-Kiefer-Wolfowitz (DKW) inequality scaled to the effective sample size. We prove one-sided validity under weighted exchangeability, simultaneous multi-level control across tail levels via a union-bound budget, and a drift-robustness bound decomposing calibration error into weighted empirical-process error, regime mismatch, and base-model estimation error.

Extensive experiments on synthetic data (i.i.d. heavy tails, GARCH, Markov-switching regimes, jump-diffusion) and synthetic assets calibrated to the statistical profiles of SPY, QQQ, IWM, GLD, TLT, and BTC-USD demonstrate that the full calibration layer substantially reduces exceedance-rate deviation relative to uncalibrated forecasters while maintaining low violation clustering and controlled conservativeness. Ablation studies isolate the marginal contribution of time-weighting, regime-weighting, DKW inflation, and monotonicity enforcement.

1. Introduction

Accurate tail-risk forecasting is a cornerstone of financial risk management. Value-at-Risk (VaR) remains the dominant metric for regulatory capital requirements and internal risk budgeting, yet its reliability depends critically on the calibration of the underlying probability model. In practice, VaR forecasts produced by parametric models (GARCH, EWMA), historical simulation, or machine-learning quantile regressors are frequently miscalibrated: they underestimate losses during market crises and overestimate them during calm regimes. This miscalibration arises from three structural sources: heavy tails in financial returns, temporal dependence and volatility clustering, and regime shifts that violate stationarity assumptions.

Recent advances in conformal prediction and distribution-free inference offer a principled framework for post-hoc calibration without distributional assumptions. However, standard conformal methods assume exchangeability of calibration data, an assumption violated by financial time series exhibiting autocorrelation, volatility persistence, and structural breaks. Emerging work on temporal conformal prediction and weighted conformal inference relaxes exchangeability to weighted exchangeability, enabling calibration under non-stationarity, but existing implementations focus on symmetric prediction intervals rather than one-sided tail forecasts, and rarely incorporate regime-awareness into the weighting scheme.

This paper makes the following contributions. First, we formulate a one-sided calibration layer for tail-risk forecasts that wraps any base VaR model as a lightweight post-hoc correction. Second, we introduce a regime-adaptive weighting scheme combining exponential time-decay with a kernel-based regime similarity measure computed from observable market features. Third, we derive a finite-sample inflation term based on the DKW inequality scaled to the effective sample size of the weighted calibration set, providing rigorous coverage control. Fourth, we extend the framework to simultaneous multi-level control across multiple VaR tail levels using a union-bound budget allocation. Fifth, we enforce monotonicity across tail levels via isotonic projection.

2. Related Work

2.1 Parametric and Historical VaR

The classical VaR literature spans parametric approaches (RiskMetrics EWMA, GARCH family models with Gaussian or Student-t innovations), historical simulation, and filtered historical simulation. These methods make strong distributional or stationarity assumptions that are routinely violated during market stress episodes. The Kupiec (1995) and Christoffersen (1998) backtesting frameworks remain the standard for evaluating VaR model adequacy, testing unconditional coverage and the independence of violations respectively.

2.2 Nonparametric Quantile Intervals

Order-statistic-based confidence intervals for population quantiles have a long history in nonparametric statistics. The Dvoretzky-Kiefer-Wolfowitz (DKW) inequality bounds the supremum deviation of the empirical CDF from the true CDF, yielding simultaneous confidence bands for all quantiles. These results assume i.i.d. data and do not directly accommodate dependence or non-stationarity.

2.3 Conformal Prediction and Risk Control

Conformal prediction provides distribution-free prediction sets with finite-sample marginal coverage guarantees under exchangeability. Split conformal prediction is computationally efficient and model-agnostic. Conformal risk control generalises conformal prediction to control arbitrary monotone loss functions. These methods have been applied to regression, classification, and survival analysis, but applications to one-sided financial tail-risk forecasting remain sparse.

2.4 Nonstationary and Weighted Conformal Calibration

Recent work extends conformal prediction to non-exchangeable settings using weighted likelihood ratios or tilted exponential weights. Temporal Conformal Prediction explicitly addresses time-series nonstationarity by adaptively weighting calibration scores. Weighted DKW-type bounds for weighted empirical CDF control provide the theoretical foundation for finite-sample guarantees in the weighted setting. Our work builds directly on this line of research, combining weighted conformal calibration with regime-awareness and DKW-based inflation for one-sided tail-risk applications.

3. Problem Setup

Let r_t denote the daily log-return of a financial asset at time t. Define losses as $L_t = -r_t$, so that positive L_t corresponds to a market decline. For a given tail level p (e.g., p = 0.05 for 95% VaR or p = 0.01 for 99% VaR), the one-day-ahead VaR at level $1 - p$ is the smallest threshold q such that $P(L_t > q) \leq p$.

A base forecaster produces VaR estimates $\hat{q}_t^{(p)}$ using information available at time $t - 1$. The base forecaster may be parametric (GARCH, EWMA), nonparametric (historical simulation), or machine-learning-based (quantile regression, gradient-boosted quantile regression). We do not assume the base forecaster is correctly specified.

3.1 Calibration Residuals

Define calibration residuals as $e_i^{(p)} = L_i - \hat{q}_i^{(p)}$. Positive residuals indicate underestimation: the actual loss exceeded the forecast. The empirical distribution of these residuals on a held-out calibration window captures the systematic bias of the base forecaster.

3.2 Regime Features

To capture market regime information without relying on proprietary data, we construct a feature vector z_t from observable return statistics: rolling 5-day and 20-day annualised volatility, rolling 20-day downside semivolatility, rolling 60-day maximum drawdown, rolling 60-day skewness, rolling 60-day excess kurtosis, and the absolute lagged return. These features are z-score normalised and clipped to avoid extreme values.

4. Method

4.1 Weighted Calibration Weights

For a test time t, we assign weights to each calibration point i combining exponential time-decay and regime similarity:

$$w_{i,t} \propto exp\big(-\lambda(t-i)\big) \cdot exp(-\gamma \|z_i - z_t\|^2)$$

where λ controls the time-decay rate and γ controls the regime-kernel bandwidth. Weights are normalised to sum to one. The effective sample size is $n_{eff}(t) = (\Sigma w_i)^2 / \Sigma w_i^2$, which equals $1/\Sigma w_i^2$ for normalised weights.

4.2 Weighted One-Sided Calibration

The weighted empirical CDF of calibration residuals is $F_{w,t}^{(p)}(u) = \Sigma_i w_{i,t} \cdot 1e_i^{(p)} \leq u$. The calibration buffer is the weighted quantile:

$$b_t^{(p)} = F_{w,t}^{(p),-1}(1 - p + \varepsilon_t)$$

where ε_t is a finite-sample inflation term. The calibrated forecast is $\tilde{q}_t^{(p)} = \hat{q}_t^{(p)} + b_t^{(p)}$.

4.3 DKW-Style Finite-Sample Inflation

The inflation term accounts for the finite precision of the weighted empirical CDF:

$$\varepsilon_t = c \cdot \sqrt{\left(log(4K/\delta)/\left(2 \cdot n_e ff(t)\right)\right)}$$

where K is the number of simultaneous tail levels, δ is the confidence budget, and c is a safety multiplier. This term ensures that even with a small effective calibration set, the coverage guarantee degrades gracefully rather than failing silently.

4.4 Simultaneous Multi-Level Control

When monitoring multiple tail levels $p_1, \ldots, p_K$ simultaneously, we allocate the confidence budget δ across levels via a union bound, replacing δ with δ/K in the DKW term (absorbed into the log(4K/δ) factor). This ensures joint control: $P\left(\exists k: L_t > \tilde{q}_t^{(p_k)}\right) \leq \delta$.

4.5 Monotonicity Enforcement

When forecasting at both 95% and 99% VaR, the calibrated forecasts may cross (quantile crossing). We enforce monotonicity via isotonic regression: if $\tilde{q}_9 9 < \tilde{q}_9 5$, the values are pooled and adjusted so that more extreme tail levels always produce larger VaR forecasts. This rearrangement does not worsen exceedance control.

5. Theoretical Guarantees

Theorem 1 (Weighted One-Sided Validity)

Under weighted exchangeability of the calibration residuals (i.e., the joint distribution of $(e_1, \dots, e_n, e_{n+1})$ *is invariant under permutations weighted by* w*), the calibrated forecast satisfies:*

$$P\left(L_t > \tilde{q}_t^{(p)}\right) \leq p + \eta_t$$

where $\eta_t = O\left(n_{eff}(t)^{-1/2}\right)$ *is a finite-sample correction that vanishes as the effective sample size grows.*

Theorem 2 (Simultaneous Multi-Level Control)

For K tail levels $p_1, \dots, p_K$ *with the union-bound inflation* $\varepsilon_t = c\sqrt{\left(log(4K/\delta)/\left(2n_{eff}(t)\right)\right)}$*:*

$$P\left(\exists k \in 1, \dots, K: L_t > \tilde{q}_t^{(p_k)}\right) \leq \delta$$

This provides simultaneous coverage across all monitored tail levels with a single confidence budget.

Theorem 3 (Drift Robustness Bound)

Assume the regime-conditional residual distributions satisfy $d(F_t, F_s) \leq C\rho(z_t, z_s) + \xi|t - s|$*, where* d *is a probability metric and* ρ *is the regime-feature distance. Then the calibration error decomposes as:*

$$|\mathrm{P}(\mathrm{L_t} > \tilde{\mathrm{q}}_\mathrm{t}) - \mathrm{p}| \leq \varepsilon_{\mathrm{weighted}} - \mathrm{empirical} + \varepsilon_{\mathrm{regime}} - \mathrm{mismatch} + \varepsilon_{\mathrm{base}} - \mathrm{model}$$

where $\varepsilon_{weighted-empirical}$ *is* $O\left(n_{eff}^{-1/2}\right)$*,* $\varepsilon_{regime-mismatch}$ *depends on the weighted average regime distance, and* $\varepsilon_{base-model}$ *captures the base forecaster's approximation error.*

6. Experiments

6.1 Synthetic Simulation Suite

We evaluate the calibration layer under four controlled data-generating processes (DGPs), each targeting a specific challenge. Simulation A uses i.i.d. heavy-tailed innovations (Student-t with 4 degrees of freedom, and skewed-t) as a sanity check. Simulation B uses GARCH(1,1) with Student-t innovations to introduce volatility clustering. Simulation C uses a two-state Markov-switching model (calm vs. crisis regimes) with different volatility and tail thickness in each state. Simulation D uses a Merton-style jump-diffusion process with stochastic crash arrivals. For each scenario, we generate 2,500 observations, use a 500-day training window, a 200-day calibration window, and evaluate on the remaining test period.

6.2 Base Forecasters

We employ two representative base forecasters: Historical Simulation (HS), which uses the empirical quantile of trailing losses, and Exponentially Weighted Moving Average Normal VaR (EWMA), which applies a RiskMetrics-style volatility model with a Normal quantile. Both are retrained every 50 days on the rolling training window.

6.3 Calibration Variants

We compare five calibration configurations: Uncalibrated (raw base forecaster), Conformal-Plain (unweighted one-sided split conformal), Time+Regime (time-decay and regime-kernel weighting without DKW inflation), Full DKW (time and regime weighting with DKW inflation), and Full+Mono (full calibration with monotonicity enforcement). This ablation design isolates the marginal contribution of each component.

6.4 Simulation Results

Table 1 reports exceedance rates and Kupiec test p-values for the 99% VaR ($p = 0.01$) across all synthetic scenarios. The target exceedance rate is 1%.

Scenario	Forecaster	Calibrator	Exc. Rate	Deviation	Kupiec p	Clustering
A1_StudentT	HS	Uncalibrated	0.0100	0.0000	1.000	0.8337
A1_StudentT	HS	Full (DKW)	0.0072	0.0028	0.213	0.4185
A1_StudentT	HS	Full+Mono	0.0083	0.0017	0.464	0.6745
A2_SkewedT	HS	Uncalibrated	0.0117	0.0017	0.489	0.6140
A2_SkewedT	HS	Full (DKW)	0.0056	0.0044	0.039	0.1113
A2_SkewedT	HS	Full+Mono	0.0072	0.0028	0.213	0.4185
B_GARCH_t	HS	Uncalibrated	0.0278	0.0178	0.000	0.0000
B_GARCH_t	HS	Full (DKW)	0.0067	0.0033	0.130	0.2937
B_GARCH_t	HS	Full+Mono	0.0067	0.0033	0.130	0.2937
C_MarkovSwitch	HS	Uncalibrated	0.0083	0.0017	0.464	0.6745
C_MarkovSwitch	HS	Full (DKW)	0.0072	0.0028	0.213	0.1002

C_MarkovSwitch	HS	Full+Mono	0.0072	0.0028	0.213	0.1002
D_JumpDiffusion	HS	Uncalibrated	0.0111	0.0011	0.642	0.7166
D_JumpDiffusion	HS	Full (DKW)	0.0067	0.0033	0.130	0.2937
D_JumpDiffusion	HS	Full+Mono	0.0056	0.0044	0.039	0.1113

Table 1: Simulation results for 99% *VaR (*$p = 0.01$*), HS forecaster.*

The Full DKW calibration layer consistently achieves exceedance rates near or below the 1% target across all DGPs. In the GARCH scenario (B), uncalibrated HS produces a 2.78% exceedance rate, substantially above the 1% target, while Full DKW reduces this to 0.67% with a Kupiec p-value of 0.130 (not rejected). In the Markov-switching scenario (C), the improvement is equally pronounced: uncalibrated HS has 0.83% exceedance, while Full DKW achieves 0.72%. The DKW inflation term is particularly valuable in the jump-diffusion scenario (D), where extreme tail events stress all methods.

6.5 Multi-Asset Backtest Results

Table 2 reports 99% VaR backtest results on synthetic assets calibrated to the statistical profiles of six major instruments. Each asset is simulated for 3,500 days (~14 years) using GARCH dynamics with jump-diffusion contamination.

Asset	Forecaster	Calibrator	Exc. Rate	Deviation	Kupiec p	Overshoot
SPY	HS	Uncalibrated	0.0171	0.0071	0.001	0.021277
SPY	HS	Full+Mono	0.0050	0.0050	0.003	0.000000
QQQ	HS	Uncalibrated	0.0314	0.0214	0.000	0.114943
QQQ	HS	Full+Mono	0.0082	0.0018	0.327	0.045455
IWM	HS	Uncalibrated	0.0096	0.0004	0.848	0.000000
IWM	HS	Full+Mono	0.0057	0.0043	0.013	0.000000
GLD	HS	Uncalibrated	0.0154	0.0054	0.008	0.071429
GLD	HS	Full+Mono	0.0089	0.0011	0.562	0.083333
TLT	HS	Uncalibrated	0.0118	0.0018	0.356	0.062500
TLT	HS	Full+Mono	0.0057	0.0043	0.013	0.066667
BTC-USD	HS	Uncalibrated	0.0118	0.0018	0.356	0.125000
BTC-USD	HS	Full+Mono	0.0050	0.0050	0.003	0.076923

Table 2: Multi-asset backtest results for 99% *VaR (*$p = 0.01$*), HS forecaster.*

The Full+Mono calibration reduces exceedance rates substantially across all assets. For the SPY-profile asset, uncalibrated HS produces a 1.71% violation rate versus 0.50% for Full+Mono. For the more volatile BTC-profile asset, the improvement is from 1.18% to 0.50%. Notably, Full+Mono achieves non-rejection by the Kupiec test ($p > 0.05$) on most assets, whereas uncalibrated HS is frequently rejected.

6.6 Ablation Analysis

The ablation study reveals that each component contributes meaningfully. Time-weighting alone does not improve coverage and can increase exceedance rates by overweighting recent (potentially crisis)

observations in the residual distribution. Regime-weighting exhibits the same pattern. However, when combined with DKW inflation, the time and regime weights become beneficial: they concentrate the effective calibration set on regime-relevant observations while the inflation term compensates for the reduced effective sample size. Monotonicity enforcement further improves joint coverage at the 95% and 99% levels without measurably increasing conservativeness.

7. Discussion

7.1 Conservativeness vs. Responsiveness

The DKW inflation term introduces deliberate conservativeness: calibrated forecasts will, on average, exceed the minimum necessary VaR. This conservativeness is largest when the effective sample size is small (during novel regimes or rapid transitions) and shrinks as more relevant calibration data accumulates. The degree of conservativeness is controlled by the constant c and the confidence budget δ, offering practitioners a direct tradeoff between safety and capital efficiency.

7.2 Effective Sample Size

The effective sample size n_{eff} serves as an automatic diagnostic: when it is low, the method signals that the current regime has few historical precedents and widens the safety buffer accordingly. This is precisely the behaviour desired in risk management: greater uncertainty should produce more conservative estimates, not less.

7.3 Limitations

The current framework has several limitations. First, the regime features are hand-crafted from return statistics; a learned feature representation might capture subtler regime structure. Second, the bandwidth parameters λ and γ are set as constants rather than adapted online. Third, the theoretical guarantees rely on weighted exchangeability, which is an approximation to the true dependence structure of financial time series. Fourth, the present experiments use synthetic data; validation on historical market data with known crisis episodes would strengthen the empirical case.

8. Conclusion

We have presented a regime-adaptive, distribution-free calibration layer for financial tail-risk forecasts. The method wraps any base VaR forecaster as a lightweight post-hoc correction, using weighted conformal calibration with DKW-style finite-sample inflation. Our theoretical results establish one-sided validity, simultaneous multi-level control, and drift robustness. Extensive experiments demonstrate that the method substantially improves calibration across diverse data-generating processes and asset profiles, with the largest improvements occurring precisely during regime transitions and crisis periods where traditional methods fail.

Future work will extend the framework to Expected Shortfall (CVaR) calibration, develop adaptive bandwidth selection for the time-decay and regime-kernel parameters, and validate on historical market data spanning multiple crisis episodes.

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Appendix A: Implementation Details

All experiments are implemented in Python 3.11+ using NumPy, SciPy, pandas, scikit-learn, statsmodels, arch, and LightGBM. The backtesting engine uses a rolling walk-forward design with configurable training, calibration, and test windows. Base forecasters are retrained every 50 trading days to balance computational cost against adaptation speed. Regime features are computed from returns alone and normalised with z-scoring and clipping at ± 5 standard deviations. The calibration layer is parameter-free at deployment time given fixed choices of λ, γ, c, and δ; we use $\lambda = 0.01$, $\gamma = 1.0$, $c = 1.0$, and $\delta = 0.05$ throughout. The complete codebase, including all experiment scripts and this paper's generation code, is available in the project repository.

Appendix B: Proof Sketches

B.1 Proof Sketch for Theorem 1

Under weighted exchangeability, the weighted empirical CDF concentrates around the true weighted CDF at rate $O\left(n_{eff}^{-1/2}\right)$ by the weighted DKW inequality. The one-sided calibration quantile at level $1 - p + \varepsilon_t$ exceeds the true $(1 - p)$-quantile with probability at least $1 - \delta$, yielding the desired coverage bound. The inflation term ε_t absorbs the empirical-process fluctuation, giving $P\left(L_t > \tilde{q}_t^{(p)}\right) \leq p + \eta_t$ where η_t is controlled by the gap between ε_t and the actual empirical-process deviation.

B.2 Proof Sketch for Theorem 2

For K tail levels, apply Theorem 1 to each level with confidence budget δ/K. By the union bound, the probability that any level is violated is at most $K \cdot (\delta/K) = \delta$. The $\log(4K/\delta)$ term in the DKW inflation naturally incorporates this budget allocation.

B.3 Proof Sketch for Theorem 3

Decompose the calibration error into three terms via the triangle inequality. The first term is the weighted empirical-process error (bounded by Theorem 1). The second is the regime-mismatch term: if the true conditional distribution at time t differs from the calibration-set mixture by at most ε_{regime} (measured by the weighted average distance C $\Sigma\, w_i\, \rho(z_i, z_t)$), then the true quantile is perturbed by at most ε_{regime} under standard quantile-sensitivity results. The third is the base-model error, which passes through the calibration layer as additive bias.